

JAMES LEINER

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EDUCATION

- Carnegie Mellon University** — Ph.D. in Statistics and Data Science 2021 – 2026 (expected)
Thesis: Advances in Human-in-the-Loop Inference
Advisor: Aaditya Ramdas
Committee: Larry Wasserman, Jing Lei, Lucas Janson, and Daniela Witten
- The University of Chicago** — M.S. in Statistics 2019 – 2021
Thesis: Novel Data Splitting Techniques for Post-Selection Inference in Poisson Regression
Advisor: Rina Barber
- The University of Chicago** — B.A. in Mathematics and Economics 2010 – 2013

PUBLICATIONS AND PREPRINTS

- James Leiner**, Aurélien Bibaut, Nathan Kallus, Aaditya Ramdas, and Koulik Khamaru. Adaptive inference for functionals of M -Estimands. *in preparation*, 2026.
- James Leiner**, Robin Dunn, and Aaditya Ramdas. Adaptive off-policy inference for M -estimators under misspecification. *arXiv:2509.14218*, 2025.
- James Leiner**, Brian Manzo, Aaditya Ramdas, and Wesley Tansey. Scalable causal structure learning via amortized conditional independence testing. *4th Conference on Causal Learning and Reasoning (CLearR)*, 2025. **(Oral)**
- James Leiner**, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Rejoinder to the Discussion of “Data fission: splitting a single data point”. *Journal of the American Statistical Association (JASA)*, 2025.
- James Leiner** and Aaditya Ramdas. Graph fission and cross-validation. *27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- James Leiner**, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Data fission: splitting a single data point. *Journal of the American Statistical Association (JASA)*, 2023. **(Discussion Paper)**

TALKS AND INVITED SESSIONS

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| IMS International Conference on Statistics and Data Science, <i>Adaptive inference for M-estimators</i> | December 2025 |
| Conference on Causal Learning and Reasoning, <i>Scalable causal structure learning</i> | May 2025 |
| Meta Central Applied Science Research Seminar, <i>Data fission and Gaussian Process regression</i> | January 2025 |
| Joint Statistical Meeting, <i>Data fission: splitting a single data point (discussion paper talk)</i> | August 2024 |
| International Seminar on Selective Inference, <i>“Inference after Latent Variable Estimation” discussant</i> | June 2022 |
| International Seminar on Selective Inference, <i>Data fission: splitting a single data point</i> | May 2022 |

RESEARCH INTERNSHIPS

- Meta**, Research Scientist Intern Summer 2026
- Developed synthetic control methods for estimating average treatment effect in advertising geo-experiments, including cross-validation for estimator/ hyperparameter selection and Gaussian process estimators
 - Redesigned donor selection using factor models and SVD to minimize estimator variance, reducing experimental-design runtime by 20× and ATT standard errors to roughly one-third of the previous approach
- Novartis Pharmaceuticals Corporation**, Biostatistics Intern Summer 2024
- Developed causal inference and off-policy evaluation methods for adaptively collected clinical trial data
 - *Publication*: Adaptive off-policy inference for M -estimators under misspecification (preprint)

- Developed causal discovery methods to identify genetic biomarkers predictive of cancer metastases
- *Publication:* Scalable causal structure learning via amortized conditional independence testing (**CLearR**)

INDUSTRY EXPERIENCE

Novantas (now Curinos), Engagement Manager

2014–2016, 2017–2019

Promoted Associate → Lead Associate → Senior Associate → Engagement Manager

(financial services management consultancy)

- Led quantitative consulting engagements (3–7 person teams) for retail and commercial banks across deposit pricing, treasury management fee pricing, stress testing, funds transfer pricing, asset/liability management, and credit risk
- Managed client and stakeholder relationships at regional and multinational banks in the U.S., Canada, and Australia
- Built SAS/Python tooling to automate development and validation of GLM, ARIMA, and Cox PH models
- Designed mortgage early warning systems for a Big Four Australian bank to identify loans at high risk of delinquency

Facebook (now Meta), Quantitative Analyst

2016

- Developed data infrastructure and analytical reporting systems supporting large-scale marketing campaigns

TEACHING EXPERIENCE

Graduate Teaching Assistant, Carnegie Mellon University

Machine Learning Capstone Course (46-983)

Fall 2025

Natural Language Processing (46-924)

Fall 2024

Statistical Machine Learning I (46-926)

Fall 2024

ABCDE of Machine Learning (36-708)

Spring 2023

Statistical Machine Learning (35-708)

Spring 2022

Modern Regression (34-601)

Fall 2021

Teaching Assistant, University of Chicago

Applied Regression Analysis (BUSN 41100), *Booth School of Business*

Fall 2020

Pre-Calculus (MATH 10500)

Fall 2011

HONORS AND AWARDS

Runner-up Poster Award, *12th International Conference on Multiple Comparisons Procedures*

Fall 2022

Departmental Scholarship, *The University of Chicago*

Fall 2019

General Honors, *The University of Chicago*

Fall 2013

Dean's List, *The University of Chicago*

2010–2013

National Merit Corporate Scholarship, *Marsh and McLennan Companies*

2010

PROFESSIONAL SERVICE

Reviewer: Journal of the American Statistical Association (3), Journal of the Royal Statistical Society Series B (3), Electronic Journal of Statistics, Biometrika, NeurIPS

COMPUTER SKILLS

Coding Python (NumPy, pandas, scikit-learn, PyTorch), R (tidyverse, ggplot2), SAS, SQL